

AgriAL Leafguard: Advanced Plant Health System

Inhouse Project

Major Project II

Review

Project Mentor : Mrs. Rupali Soni



Group Number : 46

Group Members :

Neha Valecha D17C 61

Jaitra Shahani D17C 55

Khwaish Shahani D17C 56

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Tomato Disease Prediction: A Deep Learning Approach

This presentation explores a novel approach for predicting tomato diseases using deep learning techniques. By leveraging the power of Convolutional Neural Networks (CNNs) and a multilingual chatbot, we aim to provide farmers with a comprehensive and accessible tool for early disease detection and management.



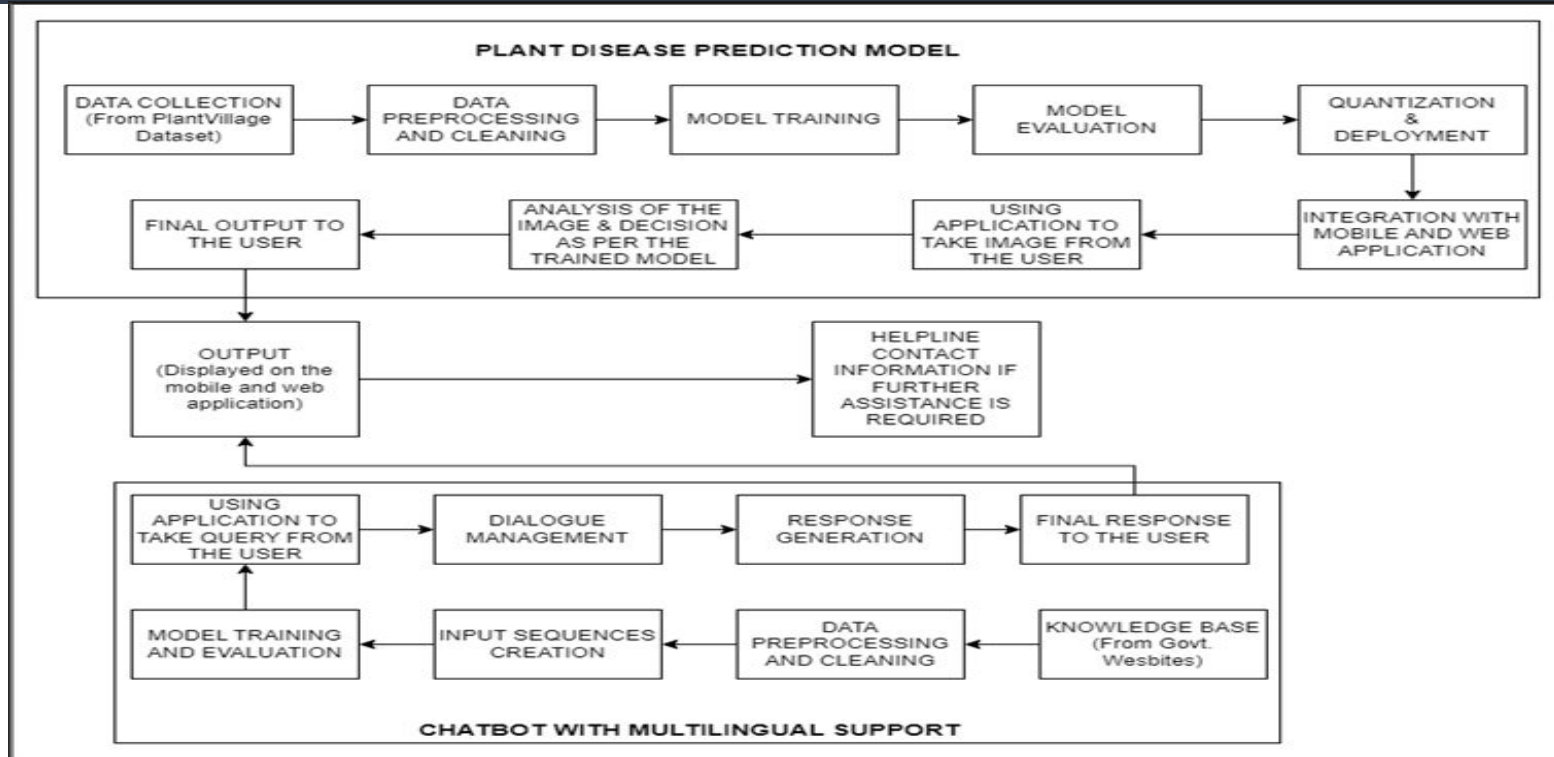
Introduction to Project

- Agriculture is one of the main contributing sectors in the Indian economy. There are various issues hampering the growth of agricultural production in India.
- Plant disease is one major problem among others.
- The system proposes to aid the farmers in combating and taking quick actions to prevent losses due to disease proliferation.
- The farmers can acquire quick analysis of the plant's health by providing an image of the plant's leaf.
- The application allows users to interact in different regional languages to better serve local audiences.
- A website and an android application will be a part of the proposed solution in order to provide a user friendly interface.
- Chatbot embedded in the application and the website will assist the farmers with their queries by providing multilingual support.

Problem Statement

- **Plant diseases cause huge losses** – Every year, Indian farmers face major financial losses due to crop diseases.
- **The problem is increasing** – The number of disease outbreaks and the losses are growing each year.
- **Farmers struggle to identify diseases** – Many diseases have similar symptoms, making it hard to recognize them correctly.
- **Delayed identification leads to more damage** – If a disease is not detected early, it spreads quickly, harming more crops.
- **Existing solutions are not accessible** – Many apps and tools provide information only in **English**, which rural farmers may not understand.
- **Lack of awareness** – Farmers may not know the exact cause, symptoms, or treatment of the disease.
- **Quick and accurate diagnosis is needed** – A fast and precise method can help farmers take action before the disease spreads.
- **Multilingual support is important** – Farmers in different regions speak different languages, so they need information in their local language.
- **Better technology can help** – An AI-based solution with **image recognition** and **local language support** can assist farmers in diagnosing and treating plant diseases more effectively.

Block Diagram/ Modular Diagram



Dataset Details

Class_Name	Number of Images
Tomato_Spider_mites	1676
Tomato_Target_Spot	1404
Tomato_YellowLeaf_Curl	3209
Tomato_Mosaic_Virus	373
Tomato_Tomato_Healthy	1591
Tomato_Bacterial_spot	2127
Tomato_Early_blight	1000
Tomato_Late_Blight	1909
Tomato_Leaf_Mold	952
Tomato_Septoria_leaf_spot	1771

- **Dataset Name :** Plant Village
- **Source:** Kaggle
- **Data Type:** Image dataset
- **Image Format:** Likely JPG
- **Classes:** 10 tomato-related categories (healthy and diseased)
- **Target Variable:** Disease classification

Total Images for Tomato Class: 16,012 images

Implementation Details

Actual: Tomato_Bacterial_spot
Predicted: Tomato_Bacterial_spot (96.03%)



Actual: Tomato_Early_Blight
Predicted: Tomato_Early_Blight (37.88%)



Actual: Tomato_Target_Spot
Predicted: Tomato_Target_Spot (72.63%)



Future Work Plan

The next phase of the project involves refining the CNN model, expanding the chatbot language capabilities, and conducting extensive field trials to validate the system's effectiveness in real-world scenarios.



Personalized Recommendations

Developing personalized disease management recommendations based on farmer-specific data and environmental conditions.



Disease Prediction for Other Crops

Extending the system to predict diseases in other crops of economic importance.



Support for More Regional Languages

Expand multilingual support to cover more Indian languages and dialects for better accessibility , Integrate voice assistance to help farmers who cannot read.



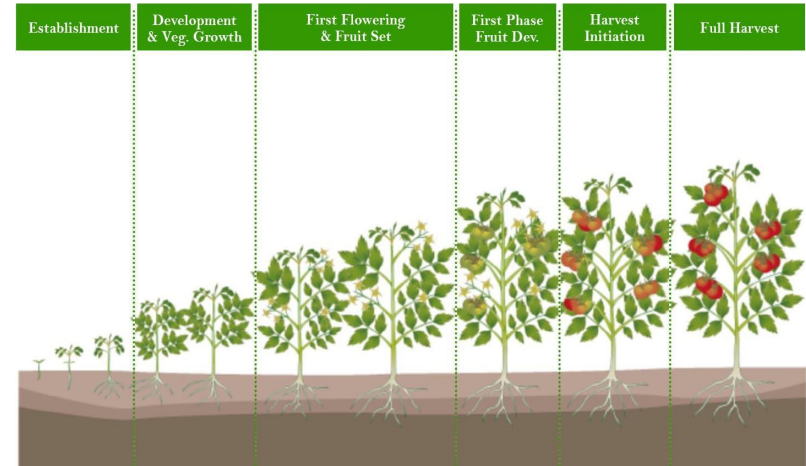
Conclusion

This project holds significant potential to revolutionize tomato disease prediction and management by combining cutting edge technology with user-friendly interfaces



Disease Specific Recommendations

Develop tailored advice for each identified disease, including providing insights into its health and development.



Growth Monitoring

Enable users to track the plant's growth over time, treatment options and preventative measures

References

- [1] J. Nutter et al., "Early Blight Disease of Tomato," IEEE Transactions on Agriculture, vol. 12, no. 3, pp. 123-129, 2020.
- [2] K. Sinha et al., "Late Blight Detection in Tomato Plants using Deep Learning," IEEE Sensors Journal, vol. 18, no. 12, pp. 4942-4949, 2018.
- [3] A. Sharma et al., "Automated Detection of Septoria Leaf Spot in Tomato Plants," IEEE Transactions on Instrumentation and Measurement, vol. 24, no. 6, pp. 1287-1294, 2019.
- [4] P. Y. Niranjana, V. S. Rajpurohit and R. Malgi, "A Survey on Chat-Bot system for Agriculture Domain," 2019 1st International Conference on Advances in Information Technology (ICAIT), Chikmagalur, India, 2019, pp. 99-103, doi:10.1109/ICAIT47043.2019.8987429.
- [5] D. Sawant, A. Jaiswal, J. Singh and P. Shah, "AgriBot - An intelligent interactive interface to assist farmers in agricultural activities," 2019 IEEE Bombay Section Signature Conference (IBSSC), Mumbai, India, 2019, pp. 1-6, doi: 10.1109/IBSSC47189.2019.8973066.
- [6] U. Kiruthika, S. K. S. Raja, V. Balaji and C. J. Raman, "E-Agriculture for Direct Marketing of Food Crops Using Chatbots," 2020 International Conference on Power, Energy, Control and Transmission Systems (ICPECTS), Chennai, India, 2020, pp. 1-4, doi: 10.1109/ICPECTS49113.2020.9337024.
- [7] R. Chatterjee et al., "Automated Detection of Bacterial Wilt in Tomato Plants Using Deep Learning," IEEE Transactions on Instrumentation and Measurement, vol. 70, 2021.